7. (a) $c = f\lambda \Rightarrow 3 \times 10^8 \text{ m s}^{-1} = f(0.02 \text{ m})$ 1M
$\therefore \quad f = 1.5 \times 10^{10}$ Hz or 15000 MHz 1A 2

(b) (i) Path difference of the diffracted waves from slits A and B to probe varies along XY. 1A
Constructive and destructive interference occur alternately to give maxima and minima. 1A 2

(ii) $BP - AP = 1\frac{1}{2}\lambda$ 1M
$BP - AP = 3$ cm $= 0.03$ m
$\therefore BP = 1.24 + 0.03 = 1.27$ m 1A 2

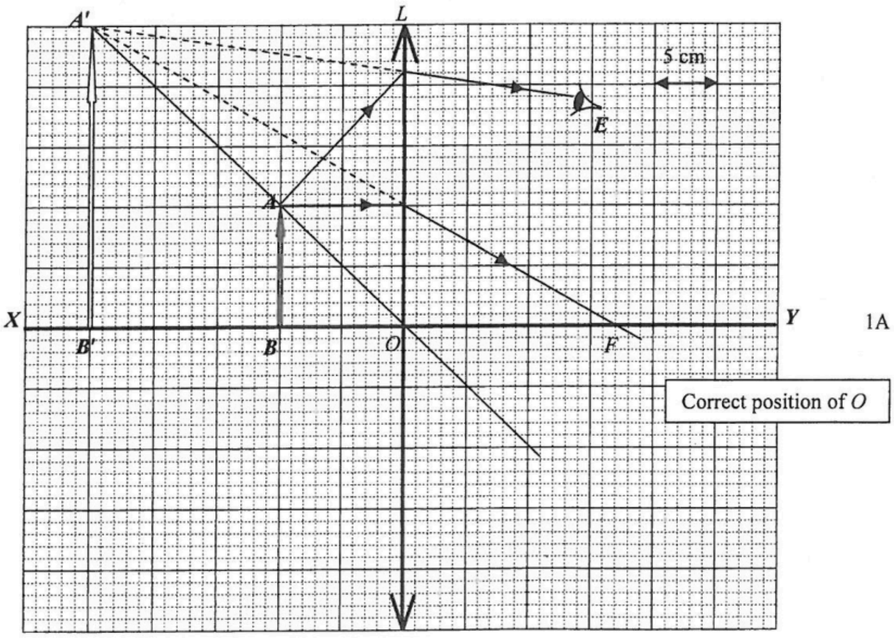
(iii) $\left.\begin{array}{l}\text{Path difference along } XY < AB \\ AB = 3 \times 2 \text{ cm} = 3\lambda \\ \therefore \text{path difference allowed} = 0\lambda, 1\lambda, 2\lambda.\end{array}\right\}$ 1M
Maximum number of maxima = 3 1A 2

(c) Radio waves with lower frequencies (will have longer wavelengths and hence) have greater diffraction effect. 1A
Radio waves by-pass small obstacles / not to be reflected from small obstacles. 1A 2

8. (a) (i) Virtual 1A 1

(ii) Convex. 1A
Only convex lens can form magnified (virtual, erect) images. 1A 2

(b) (i)



1A 1

(ii) Correct light ray to locate F. 1M
Focal length $f = 17$ cm (16.0 to 17.5 cm) 1A 2

(c) Correct ray from A' or lens to E. 1A
All correct. 1A 2

(d) Magnifying glass / glasses for long-sighted eyes / simple microscope 1A 1

9. (a)

$$k = \frac{\ln 2}{5730 \times 3.16 \times 10^7} = 3.83 \times 10^{-12} (\text{s}^{-1})$$ 1A

Activity $A = k N$

$$N = \frac{A}{k} = \frac{0.2}{3.83 \times 10^{-12}}$$ 1M

$$= 5.22 \times 10^{10}$$ 1A 3

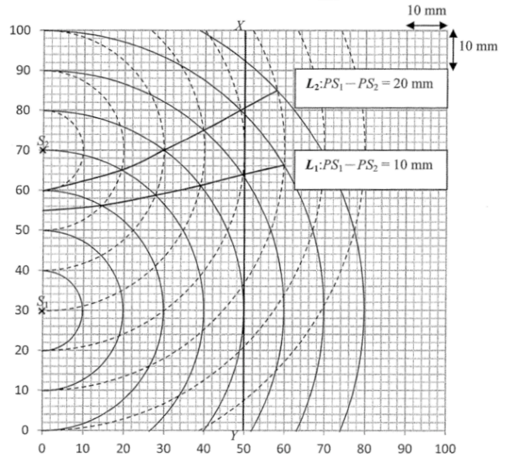
(b) No. of ^{14}C nuclei: $N_0 = 1 \times 10^{23} \times (1.3 \times 10^{-12}) = 1.3 \times 10^{11}$ 1A 1

(c)

$$kt = \ln \frac{N_0}{N}$$

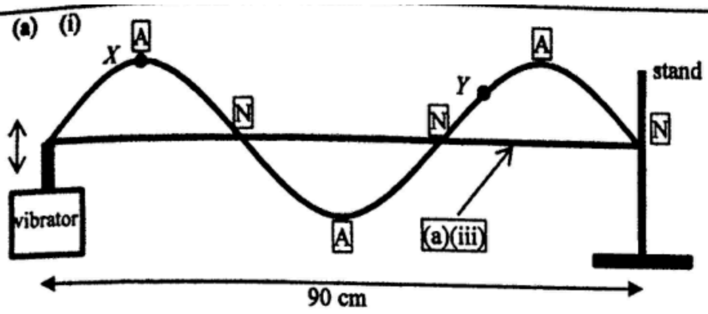
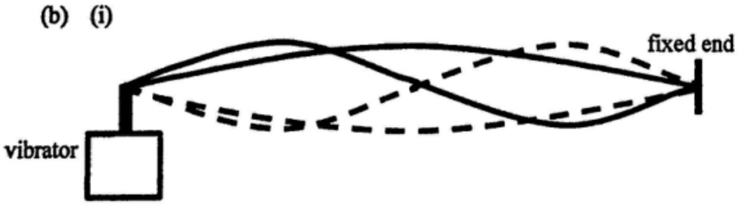
$$(3.83 \times 10^{-12})\ t = \ln \frac{1.3 \times 10^{11}}{5.2 \times 10^{10}}$$ 1M

	Marks	Remarks
7. (a) $\Delta y = \frac{\lambda D}{a}$	1M	
$= \frac{(650 \times 10^{-9}) \times 3.0}{0.325 \times 10^{-3}}$ $= 0.006$ m or 6 mm	1A	
	2	
(b) The screen is uniformly illuminated (The interference patterns exist very briefly and change rapidly such that, to human eyes, they are averaged out).	1A	NOT accept 'interference does not occur'.
The light from the LEDs is incoherent (i.e. no fixed phase relationship between the light coming out from the two LEDs).	1A	
	2	
(c) path difference $PS_1 - PS_2 = 10$ mm, L_1 correct.	1A	
path difference $PS_1 - PS_2 = 20$ mm, L_2 correct.	1A	
Constructive interference (occurs at P)	1A	
	3	



	Marks
(d) (i) $\Delta y = y_2 - y_1 = 31$ mm $- 14$ mm $= 17$ mm ± 2 mm	1A
	1
(ii) Screen has to be far away from slits or $D >> a$, (i.e. to satisfy $D >> y$ / consider y to be close to the central maximum)	1A
Or screen is too close to slits or $D >> a$ not satisfied (i.e. $D >> y$ not satisfied)	1A
Make use of small angle approximation ($\theta \approx$) $\sin\theta \approx \tan\theta$ / small angle approximation cannot be applied.	1A
	2

	Marks
8. (a) (i) $\rho = \frac{RA}{l}$ $\frac{R}{l} = \frac{\rho}{A} = \frac{2.6 \times 10^{-8}}{1.3 \times 10^{-5}}$ $= 2.0 \times 10^{-3}\ \Omega\ \text{m}^{-1}$ $= 2.0\ \Omega\ \text{km}^{-1}$ or $2.0\ \Omega$	1A
	2
(ii) The strands of transmission lines are in parallel / The cross-sectional area of cable is larger than that of each transmission line / Resistance is inversely proportional to the cross-sectional area of the cable	1A
$R_{\text{cable}} = \frac{R}{40} = 0.05\ \Omega\ \text{km}^{-1}$ or $0.05\ \Omega$ $\left(\frac{1}{R_{\text{cable}}} = \frac{1}{R} + \frac{1}{R} + \ldots + \frac{1}{R} \rightarrow \frac{1}{R_{\text{cable}}} = \frac{40}{R}\right)$	1M
	2
(iii) The resistance of the bird's body is much larger than that of a short segment of the overhead cable.	1A
Or The bird is in parallel with a short segment of the overhead cable. The potential difference across its feet is very small (very small resistance per km).	1A
Hence, negligible current flows through the bird's body.	1A
	2
(b) (i) $I = \frac{P}{V} = \frac{180 \times 10^6}{400 \times 10^3}$	1M
$= 450$ A	1A
	2
(ii) Percentage of power loss $= \frac{P_{loss}}{P_{total}} \times 100\%$	1M
$= \frac{450^2 \times 0.05 \times 10}{180 \times 10^6} \times 100\%$ $= 0.05625\ \% < 0.1\ \%$	1A
	2
(iii) (I) $N_p : N_s = V_p : V_s$ $12 : 1 = 400 : V_s$ $V_s = 33.3$ kV	1A
	1
(II) Any **ONE** of the followings: Resistance of coils + use thicker wire for the coils / Magnetisation and demagnetisation of core + use soft iron core / Induced eddy currents in core + laminated core / Flux leakage + core design	1A+1A
	2

10. (a) (i) 	2A		
		2	
(ii) Particle X vibrates with a larger amplitude than particle Y while they are vibrating in phase.	1A 1A		
		2	
(iii) Correct sketch of a straight string	1A		
		1	
(iv) Original wavelength $\lambda_1 = \frac{0.90}{1.5} = 0.60\text{ m at } f_1 = 100\text{ Hz}$ For the next stationary wave pattern, $\lambda_2 = \frac{0.90}{2} = 0.45\text{ m}$ By $v = f\lambda$, $100 \times 0.60 = f_2 \times 0.45$ $f_2 = 133.33\text{ Hz} \approx 130\text{ Hz}$	1M 1A		Or $f_2 = 100 \times \frac{4}{3}$ $= 133.33\text{ Hz} \approx 130\text{ Hz}$
		2	
(b) (i) 	1A		
		1	
(ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms.	1A 1A		
		2	
(iii) Any TWO: stationary vs travelling transverse vs longitudinal energy confined in a string vs energy being transferred through air	2A		
		2	